SQL Queries

Unicorn Database • Questions 1 – 14

01 How many customers do we have in the data?

Result	795 customers

```
SELECT
  COUNT(DISTINCT customer_id)
FROM
  customers;
```

02 What was the city with the most profit for the company in 2015?

Result	New York City — $14,753

```
SELECT
  o.shipping_city
FROM
  order_details od
  JOIN orders o ON od.order_id = o.order_id
WHERE
  EXTRACT(YEAR FROM o.order_date) = 2015
GROUP BY
  o.shipping_city
ORDER BY
  SUM(od.order_profits) DESC
LIMIT 1;
```

03 In 2015, what was the most profitable city's profit?

Result	$14,753

```
SELECT
  SUM(od.order_profits) AS total_profit
FROM
  order_details od
  JOIN orders o ON od.order_id = o.order_id
WHERE
  EXTRACT(YEAR FROM o.order_date) = 2015
GROUP BY
  o.shipping_city
ORDER BY
  total_profit DESC
LIMIT 1;
```

04 How many different cities do we have in the data?

Result	531 cities

```
SELECT
  COUNT(DISTINCT shipping_city) AS num_of_cities
FROM
  orders;
```

05 What is the most profitable city in the State of Tennessee?

Result	Lebanon

```
SELECT
  o.shipping_city
FROM
  orders o
```

```
  JOIN order_details od ON o.order_id = od.order_id
WHERE
  o.shipping_state = 'Tennessee'
GROUP BY
  o.shipping_city
ORDER BY
  SUM(od.order_profits) DESC
LIMIT 1;
```

06 What is the distribution of customer types in the data?

Result	Consumer: 410 | Corporate: 237 | Home Office: 148

```
SELECT
  customer_segment,
  COUNT(*) AS num_customers
FROM
  customers
GROUP BY
  customer_segment
ORDER BY
  num_customers DESC;
```

07 Which was the biggest order regarding sales in 2015?

Result	CA-2015-145317 — $23,660

```
SELECT
  od.order_id AS biggest_order,
  SUM(od.order_sales) AS total_sales
FROM
  order_details od
  JOIN orders o ON od.order_id = o.order_id
```

```
WHERE
  EXTRACT(YEAR FROM o.order_date) = 2015
GROUP BY
  od.order_id
ORDER BY
  total_sales DESC
LIMIT 1;
```

08	Display customer names in Consumer or Corporate segment. How many total?
Result	647 total customers

```
-- List customer names
SELECT
  customer_name
FROM
  customers
WHERE
  customer_segment IN ('Consumer', 'Corporate')
ORDER BY
  customer_name;

-- Count total
SELECT
  COUNT(*) AS total_customers
FROM
  customers
WHERE
  customer_segment IN ('Consumer', 'Corporate');
```

=> Could also use a UNION ALL to stack the queries with a CONCAT('Total Customers: ', COUNT(*))

09	Calculate the difference between the largest and smallest order quantities for product id 100.
Result	4

```
SELECT
  MAX(quantity) - MIN(quantity) AS quantity_difference
FROM
  order_details
WHERE
  product_id = 100;
```

10	Calculate the percent of products within the category Furniture.
Result	20.54%

```
SELECT
  ROUND(
    (COUNT(*) * 100.0 / (SELECT COUNT(*) FROM product)),
    2
  ) AS percent_of_total_furniture
FROM
  product
WHERE
  product_category = 'Furniture';
```

11	Display manufacturers with more than 1 product, with their product count.

```
SELECT
  product_manufacturer,
  COUNT(*) AS num_products
FROM
```

```
  product
GROUP BY
  product_manufacturer
HAVING
  COUNT(*) > 1;
```

12	**Show the product subcategory and total number of products in each subcategory.**

```
SELECT
  product_subcategory,
  COUNT(product_id) AS num_products_in_subcategory
FROM
  product
GROUP BY
  product_subcategory;
```

13	**Show subcategories ordered from most to least number of products.**

```
SELECT
  product_subcategory,
  COUNT(product_id) AS num_products_in_subcategory
FROM
  product
GROUP BY
  product_subcategory
ORDER BY
  num_products_in_subcategory DESC,
  product_subcategory;
```

14	**Show product IDs and sum of quantities where each individual sale quantity >= 100.**
Result	4 products

```
SELECT
  product_id,
  SUM(quantity) AS sum_quan
FROM
  order_details
WHERE
  quantity >= 100
GROUP BY
  product_id;
```